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Yun Zhang

Ph.D. Student

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Education

University of California, Los Angeles (UCLA) — Los Angeles, CA, USA
Ph.D. Student in Computer Science — **Sept. 2025 – Present**
Advisor: Co-advised by Prof. Jiaqi Ma and Prof. Wei Wang

Amazon Trainium Fellow (2025–2026); Graduate Dean's Scholar Award (2025–2027); Robotics: Science and Systems (RSS) Pathway Fellow 2025.

University of California, Los Angeles (UCLA) — Los Angeles, CA, USA
B.S. in Mathematics of Computer Science; B.S. in Statistics and Data Science — **Sept. 2021 – June. 2025**

Dean's Honors List (6 quarters); Member, Upsilon Pi Epsilon (CS Honor Society).

Research Experience

UCLA Mobility Lab, UCLA ScAi Lab, University of California, Los Angeles — *Los Angeles, CA, USA*
Graduate Student Researcher — ***Mar. 2024 – Present***

- Developing large-scale Vision–Language–Action (VLA) models for embodied navigation, manipulation, and autonomous driving, integrating high-level semantic reasoning with low-latency control.
- Training reinforcement-learning–based action policies in simulation to adapt to dynamic, multi-agent environments, enabling robust closed-loop control under changing scene conditions.
- Building large-scale datasets, closed-loop testing pipelines, and reinforcement-learning training environments on simulation platforms (*Isaac Lab, Isaac Sim, CARLA*) to study generalization, grounding, and failure modes in multimodal learning systems.

Vwani Roychowdhury Lab, University of California, Los Angeles — *Los Angeles, CA, USA*
Research Assistant — **Feb. 2023 – Dec. 2024**

- Contributed to PyHFO 2.0, an open-source EEG analysis framework for high-frequency oscillation (HFO) detection, by implementing and integrating CNN- and transformer-based detection modules.
- Improved computational efficiency (up to $\sim 50\times$) through algorithmic optimization, parallelization, and system-level redesign relative to a traditional MATLAB-based pipeline.
- Assessed robustness and generalization through cross-patient benchmarking and statistical evaluation.

HKU Summer Research Program, The University of Hong Kong — *Hong Kong, China*
Research Intern — **Summer 2024**

- Developed an LLM-guided multimodal brain tumor localization framework integrating heterogeneous MRI sequences (T1c, T1n, T2, FLAIR) via instruction-following, domain-specific visual encoders (BioMedCLIP), and modality fusion.
- Awarded *Best Presenter* and received a *Presidential Scholarship Ph.D. offer* through the HKU Summer Research Programme.

Selected Publications

1. Z. Huang †, **Y. Zhang** †, J. Liu, R. Song, C. Tang, and J. Ma. TIC-VLA: A Think-in-Control Vision-Language-Action Model for Robot Navigation in Dynamic Environments. *International Conference on Machine Learning (ICML)*, 2026.
2. **Y. Zhang**, Z. Zheng, J. Liu, Z. Huang, Z. Zhou, Z. Meng, T. Cai, and J. Ma. MIC-BEV: Multi-Infrastructure Camera Bird's-Eye-View Transformer with Relation-Aware Fusion for 3D Object Detection. *Submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), Best Paper Award (Third Prize) at ICCV 2025 DriveX Workshop*.

3. Z. Zheng †, **Y. Zhang** †, Z. Meng, J. Liu, X. Xia, and J. Ma. InSPE: Rapid Evaluation of Heterogeneous Multi-Modal Infrastructure Sensor Placement. *Submitted to International Conference on Intelligent Robots and Systems*, 2026.
4. Z. Meng, **Y. Zhang***, Z. Zheng, S. Z. Zhao, and J. Ma. AgentAlign: Misalignment-Adapted Multi-Agent Perception for Resilient Inter-Agent Sensor Correlations. *Journal of Intelligent Transportation Systems*, 2026.
5. T. Cai, **Y. Zhang**, Z. Zhou, Z. Huang, and J. Ma. RelMap: Enhancing Online Map Construction with Class-Aware Spatial Relation and Semantic Priors. *IEEE International Conference on Robotics & Automation (ICRA)*, 2026.
6. Zhiyu Huang, Johnson Liu, Rui Song, Zewei Zhou, Ruining Yang, **Yun Zhang**, Tianhui Cai, Hanyin Zhang, Mingxuan Gao, Valeria Xu, Jiali Chen, Yishan Shen, Yiluan Guo, Tony (Xuewei)Qi, Jiaqi Ma. nuReasoning: A Reasoning-Centric Dataset and Benchmark for Long-Tail Autonomous Driving. *Neural Information Processing Systems (NeurIPS)*, 2026.
7. X. Han, Z. Zheng, Z. Zhou, **Y. Zhang**, T. Cai, Y. Liu, H. Xiang, C. Correa-Jullian, Z. Meng, Z. Huang, L. Gao, X. Xia, and J. Ma. CDA.AI for OpenCDA: AI pathways for cooperative driving automation research. *Artificial Intelligence for Transportation*, 2025.
8. Z. Zhou, T. Cai, SZ. Zhao, **Y. Zhang**, Z. Huang, and J. Ma. AutoVLA: Vision-Language-Action Model for End-to-End Autonomous Driving with Adaptive Reasoning and Reinforcement Fine-Tuning. *Neural Information Processing Systems (NeurIPS)*, 2025.

† These authors contributed equally. * Corresponding author.

Professional Service

- **Reviewer**: *Conference on Robot Learning (CoRL)*; *IEEE Open Journal of Intelligent Transportation Systems (OJ-ITS)*; *IEEE International Conference on Intelligent Transportation Systems (ITSC)*; *IEEE Intelligent Vehicles Symposium (IV)*; *IEEE International Conference on Robotics and Automation (ICRA); International Conference on Machine Learning (ICML)*.
- **Program Committee / Organizer**
  Organizer, DriveX Workshop of *European Conference on Computer Vision (ECCV) 2026*;
  Organizer, DriveX Workshop of *IEEE Intelligent Transportation Systems Conference (ITSC) 2026*;
  Organizer, DriveX Workshop of *IEEE Intelligent Vehicles Symposium (IV) 2026*;
  Organizer, DriveX Workshop of *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2026*;
  Organizer, Tutorial on *Beyond Self-Driving: Exploring Three Levels of Driving Automation (ICCV) 2025*.

Honors & Awards

Amazon Trainium Fellowship 2025 – 2026
Awarded for research excellence in large-scale multimodal learning for embodied intelligence; supports model training on AWS Trainium hardware.

Graduate Dean's Scholar Award (GDSA), UCLA 2025 – 2027
One of UCLA's highest graduate fellowships recognizing outstanding academic and research achievements.

RSS Pathway Fellowship, Robotics: Science and Systems 2025
Selected among early-career researchers contributing to robotics foundations, embodied intelligence, and autonomous systems.

Core Developer, Winner of U.S. Department of Transportation's Intersection Safety Challenge 2025
Received a $750,000 cash prize for developing winning solutions in the U.S. DOT Smart Intersection Safety Challenge.

Best Presenter, HKU Summer Research Program Summer 2024
Awarded for outstanding research presentation on multimodal medical imaging; includes a $2770 scholarship.

Dean's Honors List, UCLA 2021 – 2024
Achieved academic distinction for six quarters during undergraduate study.

Upsilon Pi Epsilon (UPE) Honor Society Inductee 2023
Recognized for academic excellence in computer science and contributions to the computing community.

ICCV DriveX Workshop Best Paper Award (Third Prize) 2025
Awarded for contributions to infrastructure-based perception and BEV modeling (MIC-BEV project).